Math 240 Matlab Project 4

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*Type/paste Matlab commands for a particular problem/part below the correspondingly labeled line, above the next double % sign. Written responses to questions (not lines of code) should be given as comments, following a single percent symbol % NEWLINE

Problem 1

(a)

```
format rat
P = [1 1 2 4; 0 2 -1 -1; 0 0 3 0; 0 0 0 1]
Q = [1 2 0 0; 3 1 3 3; 0 0 -1 0; 1 0 4 0]
```

```
P =

     1     1     2     4
     0     2    -1    -1
     0     0     3     0
     0     0     0     1


Q =

     1     2     0     0
     3     1     3     3
     0     0    -1     0
     1     0     4     0
```

(b)

```
R = inv(Q) * P
```

```
R =

     0     0    12     1
```

```
       1/2            1/2            -5              3/2
       0              0              -3              0
      -1/6            1/2          -23/3           -11/6
```

(c)

```
t3e = [0; 0; 0; 1]
t3c = inv(Q) * t3e
```

```
t3e =

       0
       0
       0
       1


t3c =

       1
      -1/2
       0
      -5/6
```

(d)

```
pB = [0; 1; 2; 3];
pC = R * pB
```

```
pC =

      27
      -5
      -6
     -61/3
```

(e)

```
pE = P * pB
% p(t) = 17 - 3t + 6t^2 + 3t^3
```

```
pE =

      17
      -3
       6
       3
```

Problem 2

(a)

```
format short
A = [2 3 4; 5 6 7; 8 9 10]
eig(A)
```

```
A =

     2     3     4
     5     6     7
     8     9    10


ans =

   18.9499
   -0.9499
   -0.0000
```

(b) (i)

```
B = A([1,3,2], :)
eig(B)
% (ii)
C = A
C(2,:) = 2 * C(2,:)
eig(C)
% (iii)
D = A
D(1,:) = D(1,:) + (-3) * D(3,:)
eig(D)
```

```
B =

     2     3     4
     8     9    10
     5     6     7


ans =

   18.4868
   -0.4868
   -0.0000


C =

     2     3     4
```

```
     5     6     7
     8     9    10


C =

     2     3     4
    10    12    14
     8     9    10


ans =

   24.9615
   -0.9615
   -0.0000


D =

     2     3     4
     5     6     7
     8     9    10


D =

   -22   -24   -26
     5     6     7
     8     9    10


ans =

   -9.0000
    3.0000
   -0.0000
```

(c) The eigenvalues of matrices B, C, and D were different from A, but they shared one common eigenvalue: 0. This is because elementary row operations can change eigenvalues, but if the original matrix A is singular (determinant zero), the resulting matrices will also be singular.

Problem 3

(a)

```
A = [-5 3 0; -34 10 -4; 40 -12 5]
[P,D] = eig(A)


A =

    -5     3     0
```

```
   -34    10    -4
    40   -12     5


P =

    0.1799    0.1961   -0.2182
    0.5998    0.5883   -0.4364
   -0.7797   -0.7845    0.8729


D =

    5.0000         0         0
         0    4.0000         0
         0         0    1.0000
```

(b)

```
A_ver = P * D * inv(P)


A_ver =

   -5.0000    3.0000   -0.0000
  -34.0000   10.0000   -4.0000
   40.0000  -12.0000    5.0000
```

(c)

```
eigvals = diag(D)
% λ1 = 5
% λ2 = 4
% λ3 = 1
% eigvec1 = [0.1799; 0.5998; -0.7797]
% eigvec2 = [0.1961; 0.5883; -0.7845]
% eigvec3 = [-0.2182; -0.4364; 0.8729]


eigvals =

    5.0000
    4.0000
    1.0000
```

Problem 4

(a)

```
A = [3 1; 0 3]
[P,D] = eig(A)
```

```
A =

     3     1
     0     3


P =

    1.0000   -1.0000
         0    0.0000


D =

     3     0
     0     3
```

(b)

```
A_ver = P * D * inv(P)


A_ver =

     3     0
     0     3
```

(c)

```
eigA = 3
E = A - eigA * eye(2)
eigvecbase = [1; 0]


eigA =

     3


E =

     0     1
     0     0


eigvecbase =

     1
     0
```

(d) No basis for R^2 consisting of eigenvectors of A due to only one linearly independent eigenvector. This explains why diagonalization failed. A matrix is diagonalizable if it has enough linearly independent eigenvectors to form a basis.

Problem 5

(a)

```
A = [3  3  1; 3 -1 -1; 2  0  1; 0  5  2]

dot1 = dot(A(:,1), A(:,2))
dot2 = dot(A(:,3), A(:,3))


A =

     3     3     1
     3    -1    -1
     2     0     1
     0     5     2


dot1 =

     6


dot2 =

     7
```

(b)

```
AtA = A' * A


AtA =

    22     6     2
     6    35    14
     2    14     7
```

(c) The entries of A^T * A are the dot products of the columns of A.

(d) In general, the entries of A * A^T are the dot products of the rows of A.

(e)

```
AAt = A * A'
dot12 = dot(A(1,:), A(2,:))
dot33 = dot(A(3,:), A(3,:))


AAt =
```

```
    19     5     7    17
     5    11     5    -7
     7     5     5     2
    17    -7     2    29


dot12 =

     5


dot33 =

     5
```

(f)

```
Q = [1/sqrt(14) 1/sqrt(3) 5/sqrt(42); 2/sqrt(14) 1/sqrt(3) -4/sqrt(42);
     3/sqrt(14) -1/sqrt(3) 1/sqrt(42)]
QtQ = Q' * Q


Q =

    0.2673    0.5774    0.7715
    0.5345    0.5774   -0.6172
    0.8018   -0.5774    0.1543


QtQ =

    1.0000         0         0
         0    1.0000   -0.0000
         0   -0.0000    1.0000
```

(g) The computation Q^T * Q forms the identity matrix. This shows that the columns of Q are orthonormal because: - The dot product of each pair of distinct columns is zero - The dot product of each column with itself is one

(h) Since Q is a square matrix and Q^T * Q = I, it follows that Q * Q^T = I as well. So, the rows of Q are also orthonormal.

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